Online Retail II

Data Analysis & Operations Improvement Pipeline

Group Assignment · Principles of Data Science (7144)

Coventry University · NIBM Campus · MSc Data Science 2025

541,910	333,234	98.4%	£4.3M	4,191	37
Raw Records	Clean Records	Data Quality	Total Revenue	Customers	Countries

Four-Stage Analysis Pipeline

Task 1 — ETL Pipeline	Task 2 — Exploratory Data Analysis	Task 3 — Clustering Analysis	Task 4 — Forecasting Model
Extract · Validate · Transform · Load	Revenue · Products · Customers · Geography	K-Means · Hierarchical · ARI Validated	Seasonal Naive vs Gradient Boosting

Team Members

#	Name	Student ID	Task	Git Branch
01	Pasindu Ashan	COMScDS251P-001	Task 1 — ETL Pipeline	feature-pasindu-ETL-EDA
02	Anuradha Dhananjanee	COMScDS252P-005	Task 2 — Exploratory Data Analysis	feature-anuradha-EDA
03	Chamika Subashinie	COMScDS252P-001	Task 3 — Clustering Analysis	feature-chamika-cluster-analysis
04	Madushika Pramod Senevirathna	COMScDS252P-002	Task 4 — Forecasting Model	feature-madushika-forecasting-model

Task 1 — ETL Pipeline

The ETL pipeline converts the raw Online Retail II dataset into a high-integrity analytical base ready for EDA, clustering, and forecasting. A structured six-step transformation process addresses six categories of data quality deficiency, reducing 541,910 raw records to 333,234 clean records with a 98.4% composite quality score.

541,910	333,234	208,676	98.4%
Raw Records	Clean Records	Removed	Quality Score

1.1 Dataset Overview

Attribute	Detail
Data Source	Kaggle — mathchi/online-retail-ii-data-set-from-ml-repository
Time Period	01 December 2010 – 09 December 2011 (12 months)
Geography	United Kingdom (primary) · 37 countries total
Business Type	Non-store B2C and B2B online retail (giftware)
Raw Records	541,910 invoice-line records · 8 original columns
Download Method	kagglehub.dataset_download() — automatic version 3
Licence	CC BY 4.0 — Open for academic use

1.2 Data Quality Audit (Pre-Cleaning)

Quality Dimension	Column(s)	Severity	Finding
Completeness	Customer ID	HIGH — 24.93% null	135,080 rows missing primary cluster key
Consistency	Invoice	HIGH — 9,288 rows	Cancellation invoices prefixed ‘C’
Validity	Quantity / Price	MED — 13,141 rows	Negative quantities and zero/negative prices
Uniqueness	All columns	LOW — 5,231 rows	Exact duplicate records

1.3 Six-Step Cleaning Pipeline

Step	Action	Rationale	Before	After	Removed
1	Remove Cancellations	Invoices prefixed ‘C’ are financial reversals, not real sales	541,910	532,622	9,288
2	Remove Duplicates	System logging errors inflate transaction frequency counts	532,622	527,391	5,231
3	Drop Null Customer ID	Customer ID is the primary key for clustering; imputation invalid	527,391	392,733	134,658
4	Filter Qty & Price ≤0	Negative/zero values produce nonsensical	392,733	392,693	40

Step	Action	Rationale	Before	After	Removed
		Total Revenue figures			
5	IQR Outlier Removal	Tukey (1977) 1.5×IQR fence removes wholesale B2B anomalies	392,693	333,234	59,459
6	Feature Engineering	Add Year, Month, DayOfWeek, Hour, Total_Revenue for downstream tasks	333,234	333,234	0 (adds cols)
TOTAL			**541,910**	**333,234**	**208,676**

1.4 Feature Engineering

Feature	Formula	Purpose
Total_Revenue	Quantity × Price	Primary financial KPI; RFM Monetary score; forecasting target
Year	InvoiceDate.dt.year	Year-over-year trend stratification
Month	InvoiceDate.dt.month	Monthly seasonality detection; forecasting aggregation
DayOfWeek	InvoiceDate.dt.day_name()	Weekly demand pattern; operations scheduling
Hour	InvoiceDate.dt.hour	Intra-day purchasing behaviour; peak-hour analysis

1.5 Post-Load Validation

VALIDATION PASSED ✓ | Rows loaded: 333,234 · Columns: 13 · Null values: 0 · Negative Revenue: 0 · Unique Customers: 4,191 · Total Revenue: £4,299,476.18

Task 2 — Exploratory Data Analysis

The EDA spans eight structured analytical dimensions: descriptive statistics, revenue trend analysis, product analysis, customer analysis, geographic analysis, correlation analysis, anomaly detection, and a composite summary dashboard. All analysis is performed on the 333,234-record cleaned dataset produced by the ETL pipeline.

333K+	£642,985	85.6%	65%
Transactions	Nov 2011 Peak	UK Revenue Share	Revenue: Top 20% Customers

2.1 Revenue Trend Analysis

Monthly revenue remained relatively flat between January and August 2011 (approximately £230,000–£330,000/month). A sustained upward shift began from September 2011, consistent with pre-Christmas wholesale stocking patterns, reaching a peak of £642,985 in November 2011. The December 2011 dip to £175,000 reflects partial-month data rather than a structural decline.

> *Q4 Revenue Concentration: September–November accounts for over 40% of annual revenue. Revenue and order volume move together during this peak — Q4 growth is driven by both higher order frequency AND elevated average order values. This dual-driver pattern is critical for demand planning.*

2.2 Temporal Demand Patterns

Dimension	Finding	Implication
Peak Day	Thursday — £0.93M total revenue	Schedule campaigns and promotions for Thursday 10am–2pm window
Zero Activity	Saturday — near-zero revenue across full dataset	Strong signal of B2B wholesale customer base (not B2C)
Peak Hour	12:00 noon — highest single-hour revenue	Email campaigns should land before 12pm on peak days
Active Window	08:00–16:00 Monday through Friday	Logistics and customer service must be staffed during business hours

2.3 Product Analysis

PARTY BUNTING leads all products with £34,953 in revenue, followed by WHITE HANGING HEART T-LIGHT HOLDER at £31,764. Revenue-leading products are predominantly decorative and gift items. By volume, PACK OF 72 RETROSPOT CAKE CASES leads at approximately 15,000 units — nearly double the next-highest product. High-volume products are typically low-unit-price consumables, while high-revenue products are higher-priced decorative items.

> *Price Band Insight: The £1–3 price band drives the highest transaction volume AND total revenue of any price segment. It is the commercial core of the business — stock depth and availability in this band must be protected at all times.*

2.4 Customer Analysis

Customer purchase frequency is heavily right-skewed: the majority of customers place only 1–2 orders across the dataset period, while a small cohort of high-frequency buyers place 10+ orders. A Pareto analysis reveals that the top 20% of customers generate approximately 65% of total revenue — more concentrated than the classic 80/20 rule. The bottom 50% contribute less than 15% of total revenue. This directly justifies a tiered CRM strategy.

2.5 Geographic Analysis

Country / Region	Revenue Share	Strategic Priority
United Kingdom	85.6%	Dominant market — protect and grow; benchmark for operations
Germany	3.2%	Highest international market — localised marketing investment justified
France	2.8%	Strong B2B signals — basket-size and bundling campaigns
EIRE	2.5%	Geographic proximity — low-cost expansion opportunity
Remaining 33 countries	5.9%	Growth opportunity — UK concentration (85.6%) is a risk factor

2.6 Correlation & Anomaly Findings

- Quantity and revenue show a moderate positive correlation ($r = 0.513$); unit price shows weaker correlation
- Month has negligible correlation with revenue ($r = -0.027$) — aggregate monthly patterns mask intra-month variation
- Anomaly threshold: Mean + 3σ = £53.00 — approximately 2% of transactions exceed this threshold
- High-value anomalies are distributed throughout the year with Q4 concentration — represent a distinct B2B wholesale segment
- Pareto curve confirms: top 10th percentile of customers drives disproportionate share of total revenue

Task 3 — Clustering Analysis

Clustering is applied across three complementary dimensions: customer behaviour, product demand, and country-level service performance. Both K-Means and Hierarchical Clustering (Ward linkage) are applied throughout, with the Adjusted Rand Index (ARI) confirming strong method agreement in all cases.

4	3	3	0.762	0.795
Customer Segments	Product Demand Tiers	Market Performance Tiers	ARI — Customers	ARI — Countries

3.1 Customer Segmentation (RFM Clustering)

Customers are grouped using three RFM metrics: Recency (days since last purchase), Frequency (unique invoices), and Monetary (total spend). Prior to clustering, 67 outliers are removed using z-score filtering (threshold: 3), leaving 4,124 customers. Features are standardised using StandardScaler before K-Means (k=4) is applied.

K-Means Results (k=4)

Cluster	Label	Recency	Frequency	Monetary	Count	% Share
2	Loyal High-Value	12.7d	14.5	£4,158	208	5.0%
3	Active Regular	31.5d	6.6	£1,813	798	19.4%
0	Occasional Buyers	49.9d	2.2	£457	2,126	51.6%
1	Lapsed / At-Risk	250.8d	1.4	£285	992	24.1%

Hierarchical Clustering Validation (Ward, k=4)

HC Cluster	Recency	Frequency	Monetary	Count	Nearest K-Means Equivalent
0	14.5d	13.0	£3,658	309	Loyal High-Value
3	25.8d	6.2	£1,609	669	Active Regular
1	49.9d	2.3	£517	2,082	Occasional Buyers
2	241.6d	1.4	£267	1,064	Lapsed / At-Risk

ARI = 0.762 — Strong agreement. Both K-Means and Hierarchical Clustering converge on the same four customer archetypes, confirming the segmentation is robust and not an artefact of algorithm choice (threshold for strong agreement: ARI > 0.7).

Business Interpretation

Cluster	Label	Strategy
2	Loyal High-Value (5.0%)	VIP loyalty programmes, exclusive early access, reward schemes to retain and deepen engagement
3	Active Regular (19.4%)	Targeted upsell and cross-sell campaigns to encourage migration into the high-value tier
0	Occasional Buyers (51.6%)	Seasonal promotions, re-engagement emails to increase purchase frequency
1	Lapsed / At-Risk (24.1%)	Win-back campaigns with discount incentives; deprioritise if resources limited

3.2 Demand Pattern Analysis (Product-Level Clustering)

Products are clustered using an adapted product-level RFM framework: Recency (days since last ordered), Frequency (unique invoices containing the product), and Monetary (total revenue generated). K-Means with k=3 produces three demand tiers, supported by the elbow method. The silhouette score peaks at k=2 (score: 0.648) but k=3 provides three business-actionable inventory tiers.

K-Means Results (k=3)

Cluster	Label	Recency	Frequency	Monetary	Count	% Share
1	Fast-Moving	2.3d	475.8	£7,698	268	7.5%
0	Mid-Tier	14.7d	72.0	£799	2,675	74.8%
2	Slow-Moving	234.7d	13.5	£157	632	17.7%

Product Keyword & Business Action Analysis

Cluster	Top Keywords	Product Character	Business Action
[1] Fast-Moving	BAG, JUMBO, LUNCH, RETROSPOT	Practical everyday items; storage & kitchen	Maintain stock levels; prioritise in supplier agreements; prevent stockouts
[0] Mid-Tier	SET, HOLDER, DECORATION, CANDLE	Decorative & gift items; seasonal	Cross-sell with fast-movers; monitor for promotion potential
[2] Slow-Moving	GLASS, BRACELET, EASTER, GARDEN	Niche / occasion-specific items	Reduce reorder quantities; apply markdowns to clear stock

Fast-movers are functional everyday items with repeatable demand. Slow-movers are occasion-driven (Easter, Garden, jewellery) and better managed as seasonal lines rather than year-round stock.

3.3 Service Performance Clustering (Country-Level)

Country-level features are engineered and clustered to identify market performance tiers across 37 countries. Six features are used: Total Revenue, Num Orders, Num Customers, Cancellation Rate, Avg Basket Value, and Product Diversity. Both K-Means (k=3) and Hierarchical Clustering (Ward, k=3) are applied independently.

K-Means Results (k=3)

Cluster	Label	Revenue (avg)	Orders (avg)	Basket (avg)	Countries	Examples
1	Core Market	£3,680,989	15,144	£243	1	United Kingdom
2	Growth Markets	£21,653	61.7	£341	26	Germany, France, EIRE
0	Developing Markets	£5,550	7.9	£764	10	USA, Switzerland, Brazil

> *ARI = 0.795 — Strong agreement. Both methods assign all 37 countries to the same performance tiers. The high basket value in Developing Markets (£764 avg) suggests B2B buyers who spend well when they do purchase — investigate localisation barriers before scaling investment.*

3.4 Method Selection & Validation Summary

Analysis	Sil. Peak	Sil. Score	Chosen k	Rationale
Customer Segmentation	k = 2	0.518	k = 4	4 actionable CRM segments — business interpretability
Demand Patterns	k = 2	0.648	k = 3	3 inventory management tiers — elbow inflection at k=3
Service Performance	k = 2	0.852	k = 3	3 regional strategy tiers — binary split insufficient

In all three analyses the silhouette score peaks at k=2. Higher values of k were chosen to produce actionable business segments, supported in each case by the elbow method and practical business logic.

Task 4 — Forecasting Model

The forecasting pipeline predicts short-term retail demand over a 30-day horizon for two primary targets: daily revenue and daily order volume. Two competing models are evaluated on a 30-day holdout window and the better performer (by RMSE, with MAE as tiebreaker) is automatically selected per target.

£540,563	2,090	£18,019	69.67	30	374
30-Day Revenue Forecast	30-Day Order Volume	Avg Daily Revenue	Avg Daily Orders	Forecast Horizon (Days)	Training Calendar Days

4.1 Models Used

	Seasonal Naive (Baseline)	Gradient Boosting (Advanced)
Approach	Last 7 training days repeated across 30-day horizon	HistGradientBoostingRegressor — non-linear tree ensemble
Features	Weekly seasonality only — no trend learning	55 features: 7 calendar + 4 lags + 2 rolling means × 8 series
Forecasting	Simple repeat of last observed weekly pattern	Recursive multi-step — each predicted day feeds next step
Role	Minimum-performance benchmark to beat	Primary model where weekly patterns have low information
Key Setting	Predictions clipped at zero (no negative forecasts)	learning_rate=0.05, max_depth=4, max_iter=400, n_leaf=10

4.2 Feature Engineering (Gradient Boosting)

Feature Category	Details
Calendar features	Day of week, day of month, month, quarter, weekend flag, month-start/end flags (7 total)
Lag features	Values at t−1, t−7, t−14, t−28 for each of 8 modelled series (32 lag features)
Rolling statistics	7-day and 28-day rolling means for each series (16 rolling features)
Total	55 features — 7 calendar + 32 lags + 16 rolling means

4.3 Model Performance Comparison (30-Day Holdout)

Target	Seasonal Naive (Baseline)			Gradient Boosting (Advanced)		
	MAE	RMSE	WAPE	MAE	RMSE	WAPE
Daily Revenue	£5,822	£8,048	27.42%	£4,349	£5,987 ✓	20.48%
Daily Order Volume	14.70	19.06 ✓	17.84%	16.73	20.53	20.30%

Selection Rule: Lowest RMSE per target (MAE as tiebreaker). Gradient Boosting → daily_revenue | Seasonal Naive → daily_order_volume | Both models deployed simultaneously for their respective targets.

4.4 Why Mixed Results?

	GB Wins on Revenue	Seasonal Naive Wins on Orders
Signal	Higher variance; influenced by order size, product mix, pricing	Stable, highly periodic weekly cycle — low variance
Feature value	Lag & rolling features capture momentum; calendar effects beyond weekly repeat	Weekly-repeat baseline already near-optimal; little room to improve
Risk	Non-linear interactions (price × customer count) improve accuracy	Recursive GB errors compound day-over-day degrading accuracy

4.5 30-Day Forward Forecast (Dec 10, 2011 – Jan 09, 2012)

Metric	Revenue (Gradient Boosting)	Order Volume (Seasonal Naive)
30-Day Total	£540,563	2,090 orders
Average Daily	£18,019	69.67 orders/day
Peak Day	£28,758	109 orders
Business Use	Cash-flow and purchasing decisions	Staffing and logistics planning

Limitations & Future Improvements

Each analytical stage carries specific limitations that represent opportunities for future enhancement.

Area	Current Limitation	Future Improvement
ETL	38.5% of records excluded — may under-represent new/guest customers	Probabilistic identity matching via StockCode + Country + date proximity
ETL	IQR removes wholesale B2B orders — B2B patterns lost from analysis	Segment B2C/B2B prior to cleaning; maintain separate analytical pipelines
ETL	Static CSV extraction — not suitable for real-time incremental loads	Migrate to Apache Airflow or AWS Glue for scheduled delta ETL loading
EDA	No external data: promotions, holidays, macro factors not included	Enrich with UK retail calendar, Bank Holidays, promotional event flags
Clustering	Silhouette peak at k=2 overridden for business interpretability	Use gap statistic and business stakeholder validation for k selection
Clustering	Static snapshot — clusters may drift over time as behaviour changes	Rolling RFM recalculation + customer migration tracking month-over-month
Forecasting	Only 374 calendar days — recursive GB errors accumulate over 30 days	Rolling-origin or expanding-window cross-validation for robust evaluation
Forecasting	Single 30-day holdout; no uncertainty bands or prediction intervals	Add SARIMAX, Prophet, LSTM alternatives with confidence intervals

Integrated Operational Recommendations

The four analytical tasks are complementary. The table below synthesises the key strategic actions derived from their integrated findings.

Area	Recommendation	Data Basis
Customer	→ Launch VIP loyalty programme for Loyal cluster (£4,158 avg spend, n=208)	Clustering Cluster 2 — low Recency, high F&M
Customer	→ Win-back email campaigns for Lapsed segment (>250 day recency, n=992)	Clustering Cluster 1 — very high Recency, low engagement
Customer	→ Use RFM scores monthly to detect customers migrating toward at-risk status	Clustering — rolling RFM recalculation
Inventory	→ Negotiate priority supplier allocation for BAG, JUMBO, LUNCH, RETROSPOT	Demand Cluster 1 — very low Recency, very high Frequency
Inventory	→ Reduce Slow-Moving stock (Cluster 2) to free working capital	Demand Cluster 2 — 234-day avg Recency, low revenue
Inventory	→ Pre-position all inventory by end of September for November peak	EDA — £642,985 November 2011 peak month
Marketing	→ Schedule all campaigns for Thursday 10am–2pm (peak revenue window)	EDA — Thursday £0.93M, peak hour 12pm

Area	Recommendation	Data Basis
Marketing	→ Concentrate Q4 marketing budget Oct–Nov: 40%+ annual revenue concentration	EDA — Q4 seasonal pattern confirmed by forecasting
Markets	→ Basket-size and bundling campaigns across 26 Growth Market countries	Service Cluster 2 — moderate activity, low basket value
Markets	→ Investigate localisation barriers in 10 Developing Markets	Service Cluster 0 — low orders, high basket suggests B2B
Forecasting	→ Use £540K revenue forecast for cash-flow and purchasing decisions	Task 4 — Gradient Boosting, 30-day horizon
Forecasting	→ Deploy rolling forecast refresh monthly as new data accumulates	Task 4 — model improves with additional history

Conclusion

This project applied a complete data science lifecycle to the Online Retail II dataset across four coordinated analytical tasks. The ETL pipeline converted 541,910 raw records into 333,234 clean records with a 98.4% quality score. EDA identified the Q4 seasonal peak (£642,985 in November 2011), a Thursday 12pm trading window, and a Pareto-concentrated customer base. Clustering analysis produced four RFM customer segments, three product demand tiers, and three market performance tiers — all validated by Adjusted Rand Index scores above 0.7. The forecasting pipeline delivered a 30-day forward forecast of £540,563 revenue and 2,090 orders, with Gradient Boosting outperforming on revenue and Seasonal Naive outperforming on order volume. Together, these findings provide a robust, data-driven foundation for CRM strategy, inventory management, geographic expansion, and operational planning.

References

Chen, D., Shen, L., & Jin, X. (2012). Online retail II dataset. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D

McKinney, W. (2022). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media.

Provost, F., & Fawcett, T. (2013). Data science for business. O'Reilly Media.

Tukey, J. W. (1977). Exploratory data analysis. Addison-Wesley.

Wang, R. Y., & Strong, D. M. (1996). Beyond accuracy: What data quality means to data consumers. Journal of Management Information Systems, 12(4), 5–33.